

# OSI Layers - open system Interconnection.

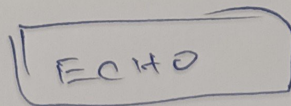
## Introduction:

- \* It's common data sharing language
- \* All network devices follow the functions of OSI layers.

## Layers:

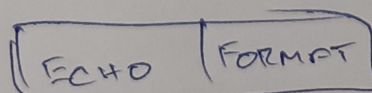
### \* Application layers:

when you type the ping Command on command prompt, the ping Command uses (ICMP) Internet control message protocol to send an echo request and receive an echo reply.



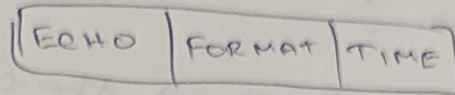
### \* Presentation layer:

Presentation layer check the echo file format.

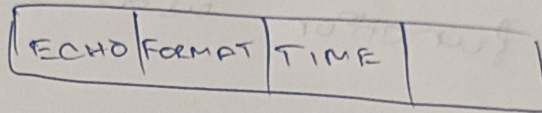




- \* Session layer: It starts, manages, and ends the communication session between two devices.

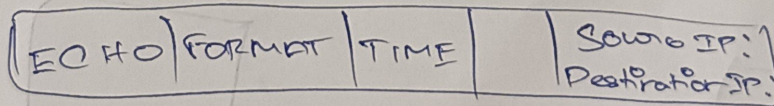


- \* Transport layer:



- \* Network layer:

- \* It's mainly handles the IP addressing
- \* It's add the source and destination IP address
- \* At this time the data becomes packet. This process is called encapsulation.



- \* Data link layer:

- \* Data link layer add the source MAC address and destination MAC address.
- \* Ethernet protocol used in this layer.
- \* we only know the source MAC address.
- \* we don't know the destination MAC address. How to get the destination MAC address? USING ARP
- \* This ARP used to find the destination MAC address with the help of destination IP address.



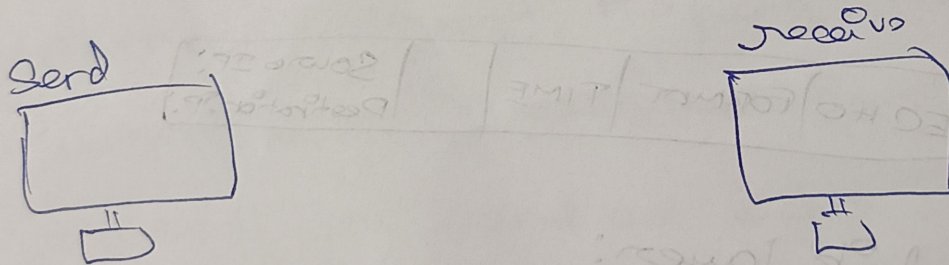
|      |        |      |     |      |
|------|--------|------|-----|------|
| ECHO | FORMAT | TIME | SIP | SMAO |
|      |        |      | DIP | DMAO |

Physical layer :

- + Physical layer is send the data the signals through cable (or) wireless
- + NIC function used in this layer

ARP table :

It store the IP address and their corresponding MAC address.



- 7 Application
- 6 Presentation
- 5 Session Presentation
- 4 Transport session
- 3 Network
- 2 Data link
- 1 Physical

1. Physical
2. Data link
3. Network
4. Transport
5. Session
6. Presentation
7. Application.